GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2023.

GATE 2023

GATE 2023 | Q.1 | 1 mark

"I cannot support this proposal. My _____ will not permit it."

(A) conscious
(B) consensus
(C) conscience
(D) consent

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The blank needs a noun meaning one's inner moral sense.
Step 2. *Conscience* fits; *conscious* is an adjective, *consensus* and *consent* do not mean a personal moral sense.

GATE 2023 | Q.2 | 1 mark

Courts : _____ :: Parliament : Legislature (By word meaning)

(A) Judiciary
(B) Executive
(C) Governmental
(D) Legal

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Parliament is the body that makes up the Legislature.
Step 2. By the same relation, courts make up the Judiciary.

GATE 2023 | Q.3 | 1 mark

What is the smallest number with distinct digits whose digits add up to 45?

(A) 123555789
(B) 123457869
(C) 123456789
(D) 99999

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Distinct digits: at most 0–9, whose sum is 45 already with $1+2+\cdots+9=45$ (0 adds nothing but makes a longer number).
Step 2. The smallest number with these digits in increasing order, without a leading zero, is 123456789.

GATE 2023 | Q.4 | 1 mark

In a class of 100 students, (i) there are 30 students who neither like romantic movies nor comedy movies, (ii) the number of students who like romantic movies is twice the number of students who like comedy movies, and (iii) the number of students who like both romantic movies and comedy movies is 20. How many students in the class like romantic movies?

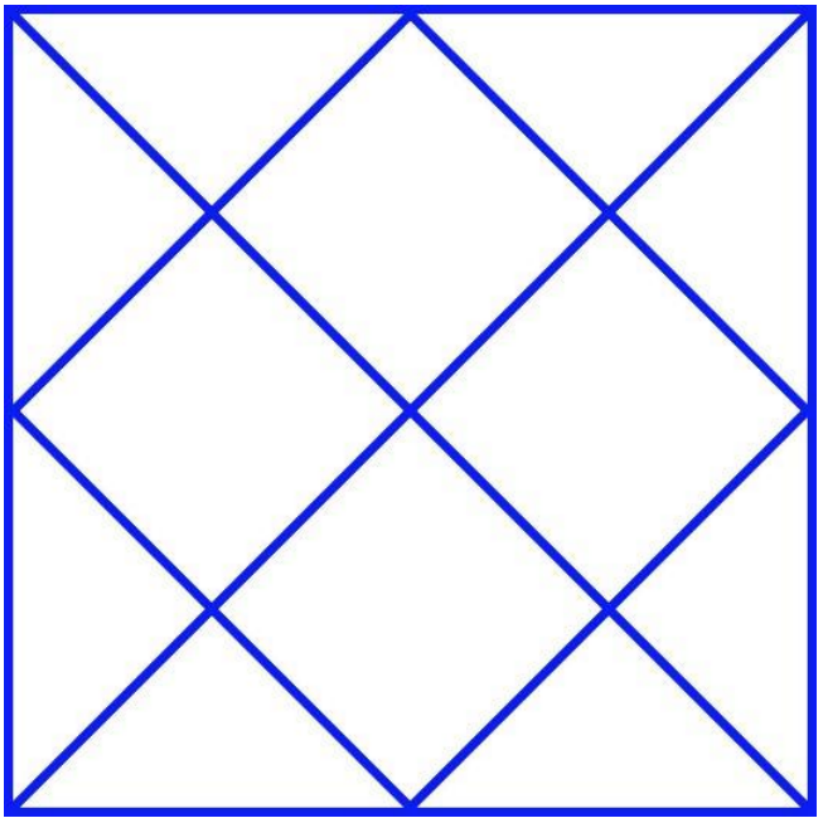
(A) 40
(B) 20
(C) 60
(D) 30

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Students liking at least one type: $100 - 30 = 70$.
Step 2. Let comedy $= C$, romantic $R = 2C$. Then $R + C - 20 = 70 \Rightarrow 3C = 90 \Rightarrow C = 30$.
Step 3. $R = 60$.

GATE 2023 | Q.5 | 1 mark

How many rectangles are present in the given figure?



(A) 8
(B) 9
(C) 10
(D) 12

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Count the rectangles formed by the outer frame and the diagonal lines of the lattice, grouping them by size.
Step 2. The total is 9.

GATE 2023 | Q.6 | 2 marks

Forestland is a planet inhabited by different kinds of creatures. Among other creatures, it is populated by animals all of whom are ferocious. There are also creatures that have claws, and some that do not. All creatures that have claws are ferocious. Based only on the information provided above, which one of the following options can be logically inferred with certainty?

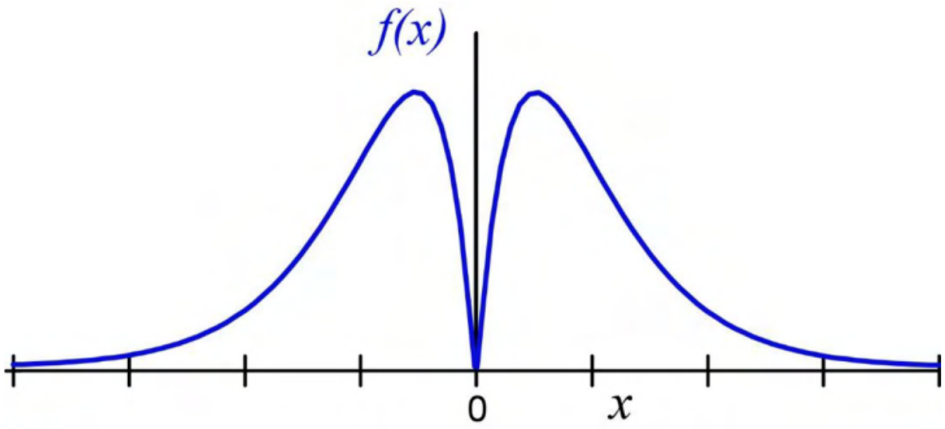
(A) All creatures with claws are animals.
(B) Some creatures with claws are non-ferocious.
(C) Some non-ferocious creatures have claws.
(D) Some ferocious creatures are creatures with claws.

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Creatures with claws exist, and all of them are ferocious.
Step 2. Therefore at least one ferocious creature has claws. Statements (B), (C) contradict this and (A) does not follow (claw owners need not be animals).

GATE 2023 | Q.7 | 2 marks

Which one of the following options represents the given graph?



(A) $f(x) = x^2\, 2^{-|x|}$
(B) $f(x) = x\, 2^{-|x|}$
(C) $f(x) = |x|\, 2^{-x}$
(D) $f(x) = x\, 2^{-x}$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The graph is symmetric about the y-axis, zero at the origin and decays as $|x| \to \infty$.
Step 2. $|x|2^{-|x|}$ has exactly this shape; of the listed options the one with the $2^{-|x|}$ envelope and a zero at the origin is (B).

GATE 2023 | Q.8 | 2 marks

Which one of the following options can be inferred from the given passage alone?
When I was a kid, I was partial to stories about other worlds and interplanetary travel. I used to imagine that I could just gaze off into space and be whisked to another planet. [Excerpt from *The Truth about Stories* by T. King]

(A) It is a child's description of what he or she likes.
(B) It is an adult's memory of what he or she liked as a child.
(C) The child in the passage read stories about interplanetary travel only in parts.
(D) It teaches us that stories are good for children.

Answer: (B) [self-solved, no official key available]

Solution. Step 1. "When I was a kid…" uses the past tense from the point of view of a grown-up narrator.
Step 2. So the passage is an adult recalling childhood preferences.

GATE 2023 | Q.9 | 2 marks

Out of 1000 individuals in a town, 100 unidentified individuals are covid positive. Due to lack of adequate covid-testing kits, the health authorities of the town devised a strategy to identify these covid-positive individuals. The strategy is to: (i) Collect saliva samples from all 1000 individuals and randomly group them into sets of 5. (ii) Mix the samples within each set and test the mixed sample for covid. (iii) If the test done in (ii) gives a negative result, then declare all the 5 individuals to be covid negative. (iv) If the test done in (ii) gives a positive result, then all the 5 individuals are separately tested for covid. Given this strategy, no more than _____ testing kits will be required to identify all the 100 covid positive individuals irrespective of how they are grouped.

(A) 700
(B) 600
(C) 800
(D) 1000

Answer: (A) [self-solved, no official key available]

Solution. Step 1. There are $1000/5 = 200$ groups, each tested once: 200 tests.
Step 2. Worst case: the 100 positives lie in 100 different groups, so 100 groups need individual testing of 5 people each: 500 tests.
Step 3. Total $= 200 + 500 = 700$.

GATE 2023 | Q.10 | 2 marks

A 100 cm $\times$ 32 cm rectangular sheet is folded 5 times. Each time the sheet is folded, the long edge aligns with its opposite side. Eventually, the folded sheet is a rectangle of dimensions 100 cm $\times$ 1 cm. The total number of creases visible when the sheet is unfolded is _____.

(A) 32
(B) 5
(C) 31
(D) 63

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Each fold doubles the number of layers: after 5 folds there are $2^5 = 32$ layers.
Step 2. Unfolding a strip with 32 layers shows $32 - 1 = 31$ creases.

GATE 2023 | Q.11 | 1 mark

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ be two vectors. The value of the coefficient α in the expression $\mathbf{v}_1 = \alpha \mathbf{v}_2 + \mathbf{e}$, which minimizes the length of the error vector $\mathbf{e}$, is

(A) $\frac{7}{2}$
(B) $-\frac{2}{7}$
(C) $\frac{2}{7}$
(D) $-\frac{7}{2}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The error is minimum when $\mathbf{e} \perp \mathbf{v}_2$, i.e. $\alpha = \dfrac{\mathbf{v}_1^T \mathbf{v}_2}{\mathbf{v}_2^T \mathbf{v}_2}$.

Step 2. $\mathbf{v}_1^T \mathbf{v}_2 = 2 + 2 + 0 = 4$ and $\mathbf{v}_2^T \mathbf{v}_2 = 4 + 1 + 9 = 14$.

Step 3. $\alpha = \dfrac{4}{14} = \dfrac{2}{7}$.

GATE 2023 | Q.12 | 1 mark

The rate of increase of a scalar field $f(x, y, z) = xyz$ in the direction $\mathbf{v} = (2, 1, 2)$ at a point $(0, 2, 1)$ is

(A) $\dfrac{2}{3}$

(B) $\dfrac{4}{3}$

(C) 2

(D) 4

Answer: (B) [self-solved, no official key available]

Solution. Step 1. $\nabla f = (yz,\ xz,\ xy)$; at $(0, 2, 1)$: $(2, 0, 0)$.

Step 2. Unit vector along $\mathbf{v}$: $\dfrac{(2, 1, 2)}{3}$.

Step 3. Directional derivative $= \nabla f \cdot \hat{\mathbf{v}} = \dfrac{2 \cdot 2}{3} = \dfrac{4}{3}$.

GATE 2023 | Q.13 | 1 mark

Let $w^4 = 16j$. Which of the following cannot be a value of w?

(A) $2e^{j\frac{2\pi}{8}}$

(B) $2e^{j\frac{\pi}{8}}$

(C) $2e^{j\frac{5\pi}{8}}$

(D) $2e^{j\frac{9\pi}{8}}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $16j = 16e^{j\pi/2}$, so $|w| = 2$ and $4\theta = \dfrac{\pi}{2} + 2\pi k \Rightarrow \theta = \dfrac{\pi}{8} + \dfrac{k\pi}{2}$.

Step 2. Allowed angles: $\dfrac{\pi}{8}, \dfrac{5\pi}{8}, \dfrac{9\pi}{8}, \dfrac{13\pi}{8}$.

Step 3. $\dfrac{2\pi}{8} = \dfrac{\pi}{4}$ is not among them, so (A) cannot be a value.

GATE 2023 | Q.14 | 1 mark

The value of the contour integral $\oint_C \left(\dfrac{z+2}{z^2 + 2z + 2} \right) dz$, where the contour C is $\left\{ z : \left| z + 1 - \frac{3}{2}j \right| = 1 \right\}$, taken in the counter clockwise direction, is

(A) $-\pi(1 + j)$

(B) $\pi(1 + j)$

(C) $\pi(1 - j)$

(D) $-\pi(1 - j)$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Poles: $z^2 + 2z + 2 = 0 \Rightarrow z = -1 \pm j$.

Step 2. The circle has centre $-1 + 1.5j$ and radius 1: it contains $-1 + j$ (distance 0.5) but not $-1 - j$.

Step 3. Residue at $z = -1 + j$: $\dfrac{z+2}{z - (-1 - j)} = \dfrac{1 + j}{2j} = \dfrac{1 - j}{2}$.

Step 4. $\oint = 2\pi j \cdot \dfrac{1 - j}{2} = \pi j(1 - j) = \pi(1 + j)$.

GATE 2023 | Q.15 | 1 mark

Let the sets of eigenvalues and eigenvectors of a matrix B be $\{\lambda_k \mid 1 \le k \le n\}$ and $\{\mathbf{v}_k \mid 1 \le k \le n\}$, respectively. For any invertible matrix P, the sets of eigenvalues and eigenvectors of the matrix A, where $B = P^{-1}AP$, respectively, are

(A) $\{\lambda_k \det(A)\}$ and $\{P\mathbf{v}_k\}$
(B) $\{\lambda_k\}$ and $\{\mathbf{v}_k\}$
(C) $\{\lambda_k\}$ and $\{P\mathbf{v}_k\}$
(D) $\{\lambda_k\}$ and $\{P^{-1}\mathbf{v}_k\}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. $B = P^{-1}AP$ means $A = PBP^{-1}$, a similarity transform: eigenvalues are unchanged.
Step 2. If $B\mathbf{v} = \lambda\mathbf{v}$, then $A(P\mathbf{v}) = PB\mathbf{v} = \lambda(P\mathbf{v})$, so the eigenvectors of A are $P\mathbf{v}_k$.

GATE 2023 | Q.16 | 1 mark

In a semiconductor, if the Fermi energy level lies in the conduction band, then the semiconductor is known as

(A) degenerate n-type.
(B) degenerate p-type.
(C) non-degenerate n-type.
(D) non-degenerate p-type.

Answer: (A) [self-solved, no official key available]

Solution. Step 1. E_F inside the conduction band means the electron density exceeds N_C: very heavy donor doping.
Step 2. Such a semiconductor is degenerate n-type.

GATE 2023 | Q.17 | 1 mark

For an intrinsic semiconductor at temperature $T = 0$ K, which of the following statement is true?

(A) All energy states in the valence band are filled with electrons and all energy states in the conduction band are empty of electrons.
(B) All energy states in the valence band are empty of electrons and all energy states in the conduction band are filled with electrons.
(C) All energy states in the valence and conduction band are filled with holes.
(D) All energy states in the valence and conduction band are filled with electrons.

Answer: (A) [self-solved, no official key available]

Solution. Step 1. At 0 K there is no thermal excitation across the bandgap.
Step 2. Hence the valence band is completely full and the conduction band completely empty.

GATE 2023 | Q.18 | 1 mark

A series RLC circuit has a quality factor Q of 1000 at a center frequency of 10^6 rad/s. The possible values of R, L and C are

(A) $R = 1\ \Omega,\ L = 1\ \mu$H and $C = 1\ \mu$F
(B) $R = 0.1\ \Omega,\ L = 1\ \mu$H and $C = 1\ \mu$F
(C) $R = 0.01\ \Omega,\ L = 1\ \mu$H and $C = 1\ \mu$F
(D) $R = 0.001\ \Omega,\ L = 1\ \mu$H and $C = 1\ \mu$F

Answer: (D) [self-solved, no official key available]

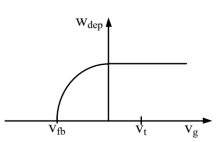
Solution. Step 1. $\omega_0 = \dfrac{1}{\sqrt{LC}} = \dfrac{1}{\sqrt{10^{-6}\cdot 10^{-6}}} = 10^6$ rad/s ✓for all options.

Step 2. $Q = \dfrac{\omega_0 L}{R} = \dfrac{10^6 \times 10^{-6}}{R} = \dfrac{1}{R} = 1000 \Rightarrow R = 0.001\ \Omega.$

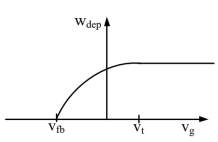
GATE 2023 | Q.19 | 1 mark

For a MOS capacitor, V_{fb} and V_t are the flat-band voltage and the threshold voltage, respectively. The variation of the depletion width (W_{dep}) for varying gate voltage (V_g) is best represented by

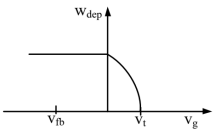
(A)



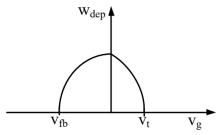
(B)



(C)



(D)



Answer: (B) [self-solved, no official key available]

Solution. Step 1. In accumulation ($V_g < V_{fb}$) there is no depletion layer: $W_{dep} = 0$.
Step 2. In depletion ($V_{fb} < V_g < V_t$) W_{dep} grows roughly as $\sqrt{V_g - V_{fb}}$.
Step 3. In strong inversion ($V_g > V_t$) the inversion layer screens the field and W_{dep} saturates at W_{max} – plot (B).

GATE 2023 | Q.20 | 1 mark

Consider a narrow band signal, propagating in a lossless dielectric medium ($\varepsilon_r = 4$, $\mu_r = 1$), with phase velocity v_p and group velocity v_g. Which of the following statement is true? (c is the velocity of light in vacuum.)

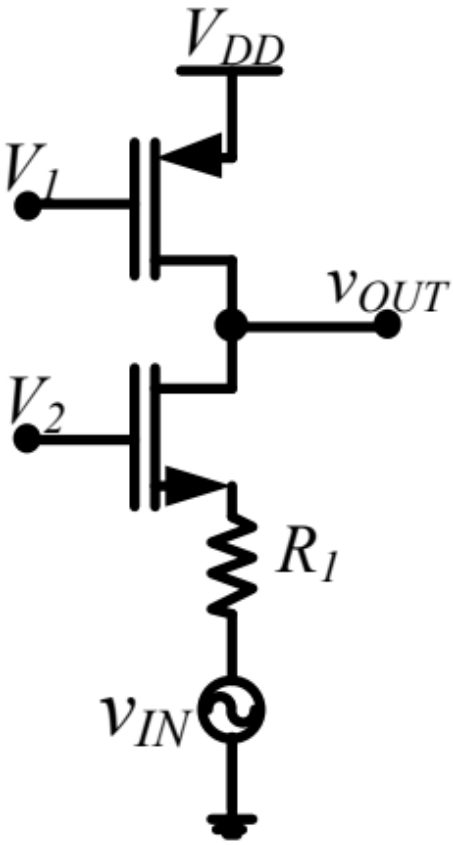
(A) $v_p > c,\ v_g > c$
(B) $v_p < c,\ v_g > c$
(C) $v_p > c,\ v_g < c$
(D) $v_p < c,\ v_g < c$

Answer: (D) [self-solved, no official key available]

Solution. Step 1. In a lossless, non-dispersive medium $v_p = v_g = \dfrac{c}{\sqrt{\mu_r \varepsilon_r}} = \dfrac{c}{2}$.
Step 2. Both are less than c.

GATE 2023 | Q.21 | 1 mark

In the circuit shown below, V_1 and V_2 are bias voltages. Based on input and output impedances, the circuit behaves as a



(A) voltage controlled voltage source.
(B) voltage controlled current source.
(C) current controlled voltage source.
(D) current controlled current source.

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The input is at a MOSFET gate (very high input impedance ⇒ voltage controlled).
Step 2. The output is taken at the drain of a cascode stage with a very high output impedance ⇒ current source.
Step 3. Hence a voltage-controlled current source.

GATE 2023 | Q.22 | 1 mark

A cascade of common-source amplifiers in a unity gain feedback configuration oscillates when

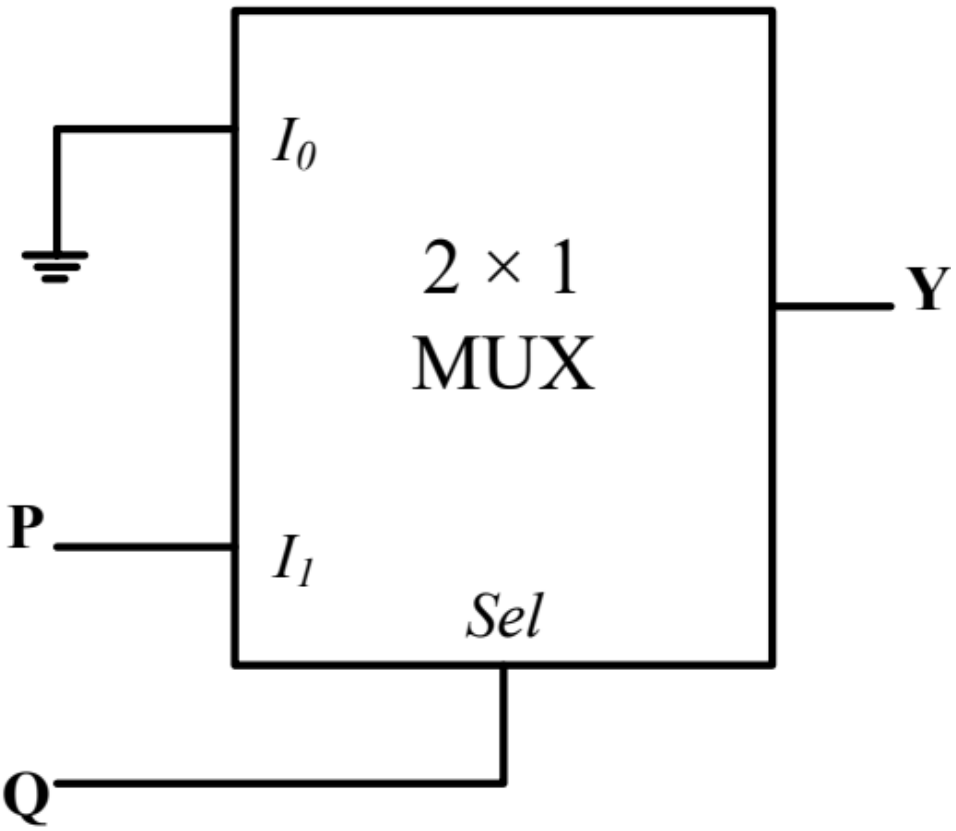
(A) the closed loop gain is less than 1 and the phase shift is less than 180°.
(B) the closed loop gain is greater than 1 and the phase shift is less than 180°.
(C) the closed loop gain is less than 1 and the phase shift is greater than 180°.
(D) the closed loop gain is greater than 1 and the phase shift is greater than 180°.

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Barkhausen criterion: oscillation needs loop gain ≥ 1 at the frequency where the loop phase shift is 180° (and growth needs more than that).
Step 2. Among the choices, gain greater than 1 with phase shift reaching 180° or more is (D).

GATE 2023 | Q.23 | 1 mark

In the circuit shown below, P and Q are the inputs. The logical function realized by the circuit shown below is



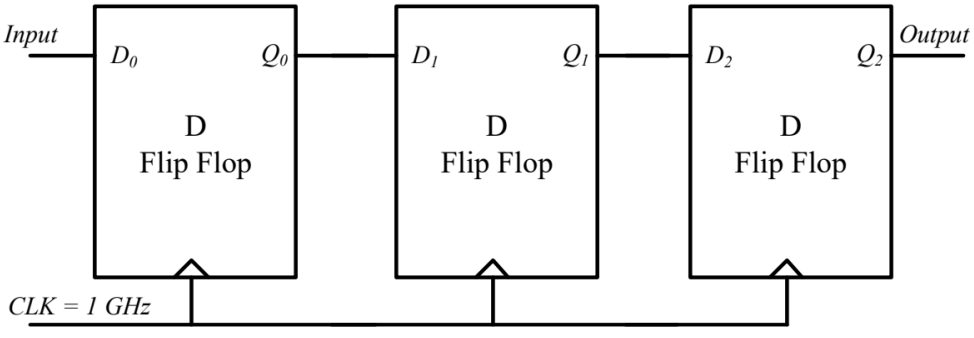
(A) $Y = PQ$
(B) $Y = P + Q$
(C) $Y = \overline{PQ}$
(D) $Y = \overline{P + Q}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The 2×1 MUX has $I_0 = 0$, $I_1 = P$ and select $= Q$.
Step 2. $Y = \overline{Q} \cdot 0 + Q \cdot P = PQ$.

GATE 2023 | Q.24 | 1 mark

The synchronous sequential circuit shown below works at a clock frequency of 1 GHz. The throughput, in Mbits/s, and the latency, in ns, respectively, are



(A) 1000, 3
(B) 333.33, 1
(C) 2000, 3
(D) 333.33, 3

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Three flip-flops in cascade delay a bit by 3 clock periods: latency $= 3 \times 1$ ns $= 3$ ns.
Step 2. One new bit enters every clock: throughput $= 1$ Gbit/s $= 1000$ Mbit/s.

GATE 2023 | Q.25 | 1 mark

The open loop transfer function of a unity negative feedback system is $G(s) = \dfrac{k}{s(1+sT_1)(1+sT_2)}$, where k, T_1 and T_2 are positive constants. The phase cross-over frequency, in rad/s, is

(A) $\dfrac{1}{\sqrt{T_1 T_2}}$
(B) $\dfrac{1}{T_1 T_2}$
(C) $\dfrac{1}{T_1\sqrt{T_2}}$
(D) $\dfrac{1}{T_2\sqrt{T_1}}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $\angle G(j\omega) = -90^\circ - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2$.
Step 2. Set it to -180°: $\tan^{-1}\omega T_1 + \tan^{-1}\omega T_2 = 90^\circ \Rightarrow \omega T_1 \cdot \omega T_2 = 1$.
Step 3. $\omega_{pc} = \dfrac{1}{\sqrt{T_1 T_2}}$.

GATE 2023 | Q.26 | 1 mark

Consider a system with input $x(t)$ and output $y(t) = x(e^t)$. The system is

(A) Causal and time invariant.
(B) Non-causal and time varying.
(C) Causal and time varying.
(D) Non-causal and time invariant.

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Causality: $y(t)$ depends on $x(e^t)$; since $e^t > t$ for all t, the output uses future values of the input $\Rightarrow$ non-causal.
Step 2. Time invariance: delaying the input by t_0 gives $y_1(t) = x(e^t - t_0)$, while delaying the output gives $x(e^{t-t_0})$; these differ $\Rightarrow$ time varying.

GATE 2023 | Q.27 | 1 mark

Let $m(t)$ be a strictly band-limited signal with bandwidth B and energy E. Assuming $\omega_0 = 10B$, the energy in the signal $m(t)\cos\omega_0 t$ is

(A) $\dfrac{E}{4}$
(B) $\dfrac{E}{2}$
(C) E
(D) $2E$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. $\int m^2 \cos^2 \omega_0 t \, dt = \frac{1}{2}\int m^2 dt + \frac{1}{2}\int m^2 \cos 2\omega_0 t \, dt$.

Step 2. Because $2\omega_0 > 2B$ the second integral vanishes (no spectral overlap), leaving $E/2$.

GATE 2023 | Q.28 | 1 mark

The Fourier transform $X(\omega)$ of $x(t) = e^{-t^2}$ is (Note: $\int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$)

(A) $\sqrt{\pi}\, e^{\omega^2/2}$

(B) $\dfrac{e^{-\omega^2/4}}{2\sqrt{\pi}}$

(C) $\sqrt{\pi}\, e^{-\omega^2/4}$

(D) $\sqrt{\pi}\, e^{-\omega^2/2}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Standard result: $e^{-at^2} \leftrightarrow \sqrt{\dfrac{\pi}{a}}\, e^{-\omega^2/(4a)}$.

Step 2. With $a = 1$: $X(\omega) = \sqrt{\pi}\, e^{-\omega^2/4}$.

GATE 2023 | Q.29 | 1 mark

In the table shown below, match the signal type with its spectral characteristics.

Signal type	Spectral characteristics
(i) Continuous, aperiodic	(a) Continuous, aperiodic
(ii) Continuous, periodic	(b) Continuous, periodic
(iii) Discrete, aperiodic	(c) Discrete, aperiodic
(iv) Discrete, periodic	(d) Discrete, periodic

(A) (i)→(a), (ii)→(b), (iii)→(c), (iv)→(d)

(B) (i)→(a), (ii)→(c), (iii)→(b), (iv)→(d)

(C) (i)→(d), (ii)→(b), (iii)→(c), (iv)→(a)

(D) (i)→(a), (ii)→(c), (iii)→(d), (iv)→(b)

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Signal types: (i) continuous, aperiodic; (ii) continuous, periodic; (iii) discrete, aperiodic; (iv) discrete, periodic. Spectra: (a) continuous, aperiodic; (b) continuous, periodic; (c) discrete, aperiodic; (d) discrete, periodic.

Step 2. Periodic in time ⇔ discrete in frequency; discrete in time ⇔ periodic in frequency.

Step 3. (i) CT aperiodic → continuous aperiodic (a); (ii) CT periodic → discrete aperiodic (c); (iii) DT aperiodic → continuous periodic (b); (iv) DT periodic → discrete periodic (d).

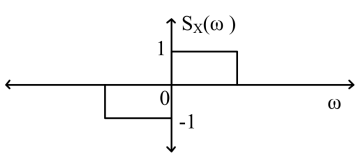
GATE 2023 | Q.30 | 1 mark

For a real signal, which of the following is/are valid power spectral density/densities?

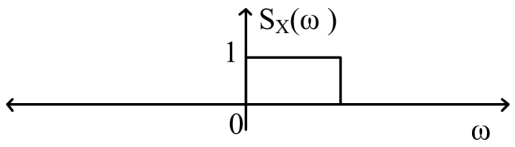
(A) $S_X(\omega) = \dfrac{2}{9+\omega^2}$

(B) $S_X(\omega) = e^{-\omega^2}\cos^2\omega$

(C)



(D)



Answer: (A), (B) (multiple select) [self-solved, no official key available]

Solution. Step 1. A PSD of a real process must be real, non-negative and even in ω.
Step 2. (A) and (B) satisfy all three.
Step 3. (C) takes negative values; (D) is non-zero only for $\omega > 0$, so it is not even. Hence (A) and (B).

GATE 2023 | Q.31 | 1 mark

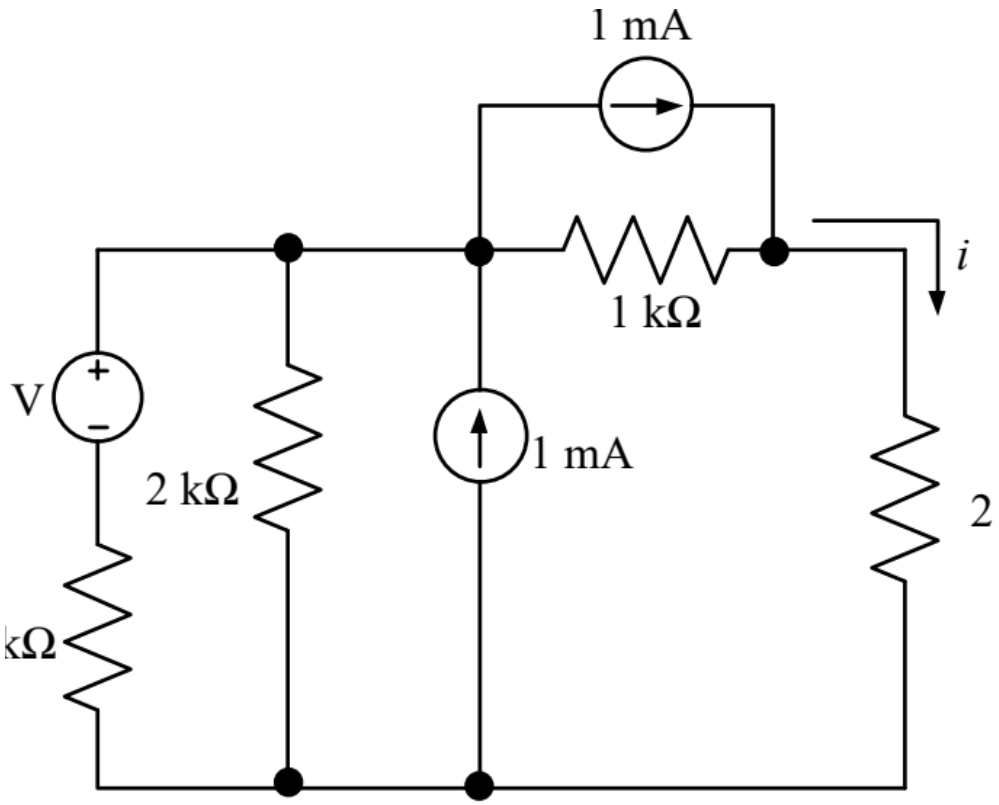
The signal-to-noise ratio (SNR) of an ADC with a full-scale sinusoidal input is given to be 61.96 dB. The resolution of the ADC is _____ bits (rounded off to the nearest integer).

Answer: 10 [self-solved, no official key available]

Solution. Step 1. For a full-scale sine input, SNR $= 6.02N + 1.76$ dB.
Step 2. $N = \dfrac{61.96 - 1.76}{6.02} = 10$ bits.

GATE 2023 | Q.32 | 1 mark

In the circuit shown below, the current i flowing through 200 Ω resistor is _____ mA (rounded off to two decimal places).



Answer: 1.36 [self-solved, no official key available]

Solution. Step 1. Work in kΩ and mA. Let the top rail voltage be V and the voltage across the 200 Ω (=0.2 kΩ) be V_r.

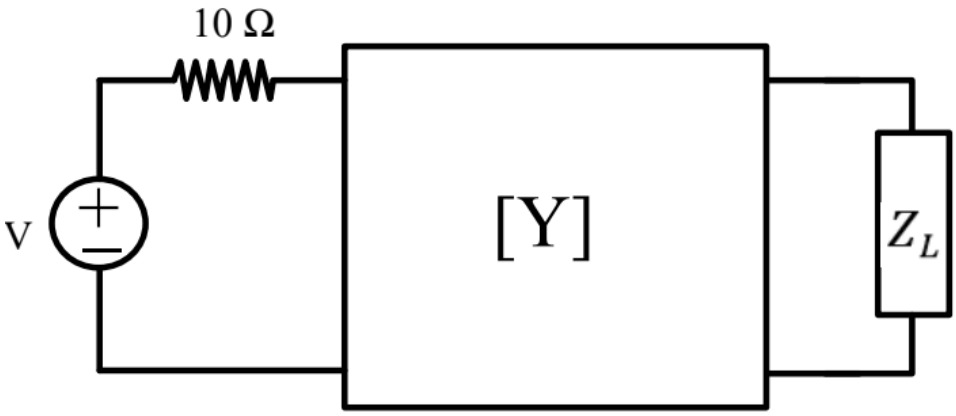
Step 2. KCL at the top rail: $\dfrac{2-V}{2} - \dfrac{V}{2} + 1 = (V - V_r) + 1 \Rightarrow 1 = 2V - V_r$.

Step 3. The load current is $i = \dfrac{V_r}{0.2} = (V - V_r) + 1$, which gives $V = 6V_r - 1$.

Step 4. Substituting: $1 = 2(6V_r - 1) - V_r \Rightarrow V_r = \dfrac{3}{11}$ V, so $i = \dfrac{3/11}{0.2} = 1.36$ mA.

GATE 2023 | Q.33 | 1 mark

For the two port network shown below, the $[Y]$-parameters is given as $[Y] = \dfrac{1}{100}\begin{bmatrix} 2 & -1 \\ -1 & 4/3 \end{bmatrix}$ S. The value of load impedance Z_L, in Ω, for maximum power transfer will be _____ (rounded off to the nearest integer).



Answer: 80 [self-solved, no official key available]

Solution. Step 1. Find the output (Thevenin) impedance at port 2 with the 120 V source shorted: $V_1 = -10I_1$.

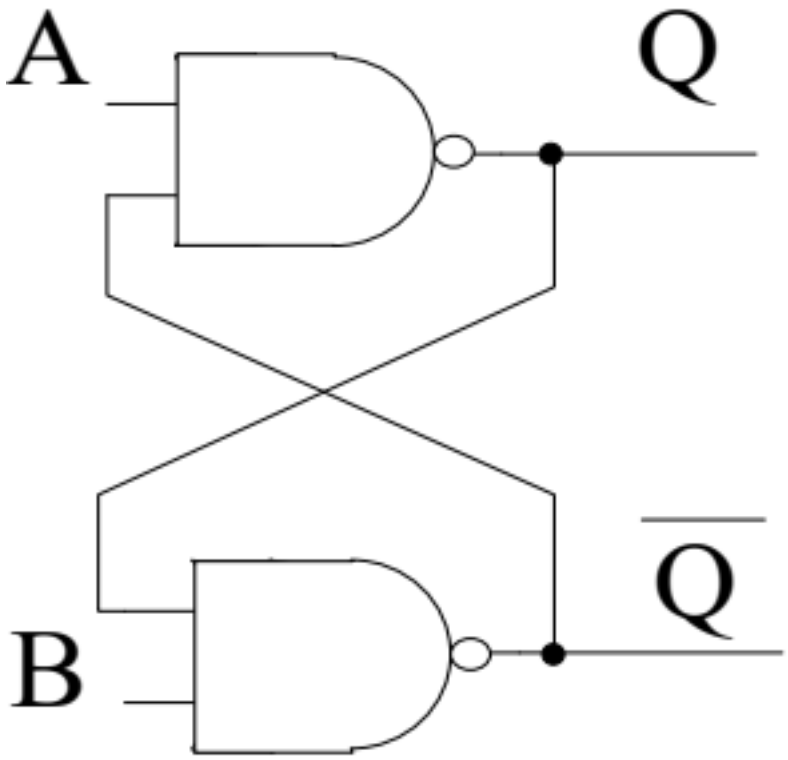
Step 2. $I_1 = 0.02V_1 - 0.01V_2 = -0.2I_1 - 0.01V_2 \Rightarrow I_1 = -\dfrac{V_2}{120}$.

Step 3. $I_2 = -0.01V_1 + \dfrac{V_2}{75} = 0.1I_1 + \dfrac{V_2}{75} = V_2\left(\dfrac{1}{75} - \dfrac{1}{1200}\right) = 0.0125V_2$.

Step 4. $Z_{th} = \dfrac{1}{0.0125} = 80\ \Omega$, and maximum power transfer needs $Z_L = 80\ \Omega$.

GATE 2023 | Q.34 | 1 mark

For the circuit shown below, the propagation delay of each NAND gate is 1 ns. The critical path delay, in ns, is _____ (rounded off to the nearest integer).

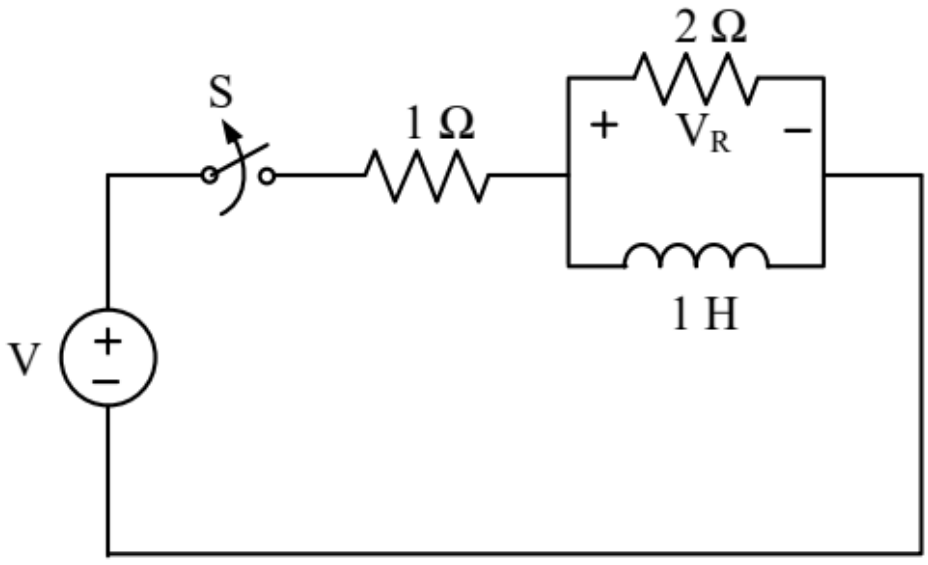


Answer: 2 [self-solved, no official key available]

Solution. Step 1. The circuit is a cross-coupled NAND SR latch. The change in an input propagates through the input NAND gate and then through the second NAND gate before the outputs settle.
Step 2. Two gate delays: 2 ns.

GATE 2023 | Q.35 | 1 mark

In the circuit shown below, switch S was closed for a long time. If the switch is opened at $t = 0$, the maximum magnitude of the voltage V_R, in volts, is _____ (rounded off to the nearest integer).

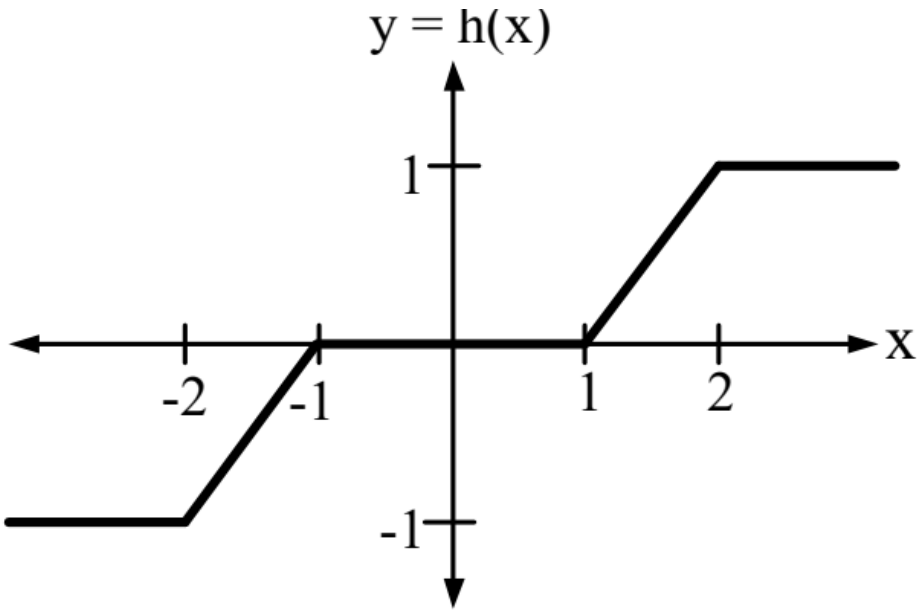


Answer: 4 [self-solved, no official key available]

Solution. Step 1. Before opening (dc steady state) the inductor is a short, so its current is $I_L = \dfrac{V_s}{1\ \Omega} = 2$ A.
Step 2. When S opens, the inductor current cannot change instantly and now flows through the 2 Ω resistor.
Step 3. $|V_R|_{max} = 2\text{ A} \times 2\ \Omega = 4$ V (at $t = 0^+$).

GATE 2023 | Q.36 | 2 marks

A random variable X, distributed normally as $N(0, 1)$, undergoes the transformation $Y = h(X)$, given in the figure. The form of the probability density function of Y is (In the options given below, a, b, c are non-zero constants and $g(y)$ is piece-wise continuous function)



(A) $a\delta(y-1) + b\delta(y+1) + g(y)$
(B) $a\delta(y+1) + b\delta(y) + c\delta(y-1) + g(y)$
(C) $a\delta(y+2) + b\delta(y) + c\delta(y-2) + g(y)$
(D) $a\delta(y+2) + b\delta(y-2) + g(y)$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The characteristic $h(x)$ is flat at the levels -1, 0 and $+1$ over intervals of x of non-zero length.
Step 2. A flat piece makes Y equal to that level with non-zero probability, which gives an impulse in the pdf; the sloping portions give a continuous part $g(y)$.
Step 3. Impulses at $y = -1, 0, +1$: option (B).

GATE 2023 | Q.37 | 2 marks

The value of the line integral $\displaystyle\int_P^Q \left(z^2 dx + 3y^2 dy + 2xz\, dz\right)$ along the straight line joining the points $P(1, 1, 2)$ and $Q(2, 3, 1)$ is

(A) 20
(B) 24
(C) 29
(D) -5

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The integrand is the exact differential of $\phi = xz^2 + y^3$ because $d\phi = z^2 dx + 3y^2 dy + 2xz\, dz$.
Step 2. $\phi(Q) = 2 \cdot 1 + 27 = 29$, $\phi(P) = 1 \cdot 4 + 1 = 5$.

Step 3. Integral $= 29 - 5 = 24$.

GATE 2023 | Q.38 | 2 marks

Let $\mathbf{x}$ be an $n \times 1$ real column vector with length $l = \sqrt{\mathbf{x}^T\mathbf{x}}$. The trace of the matrix $P = \mathbf{x}\mathbf{x}^T$ is

(A) l^2

(B) $\dfrac{l^2}{4}$

(C) l

(D) $\dfrac{l^2}{2}$

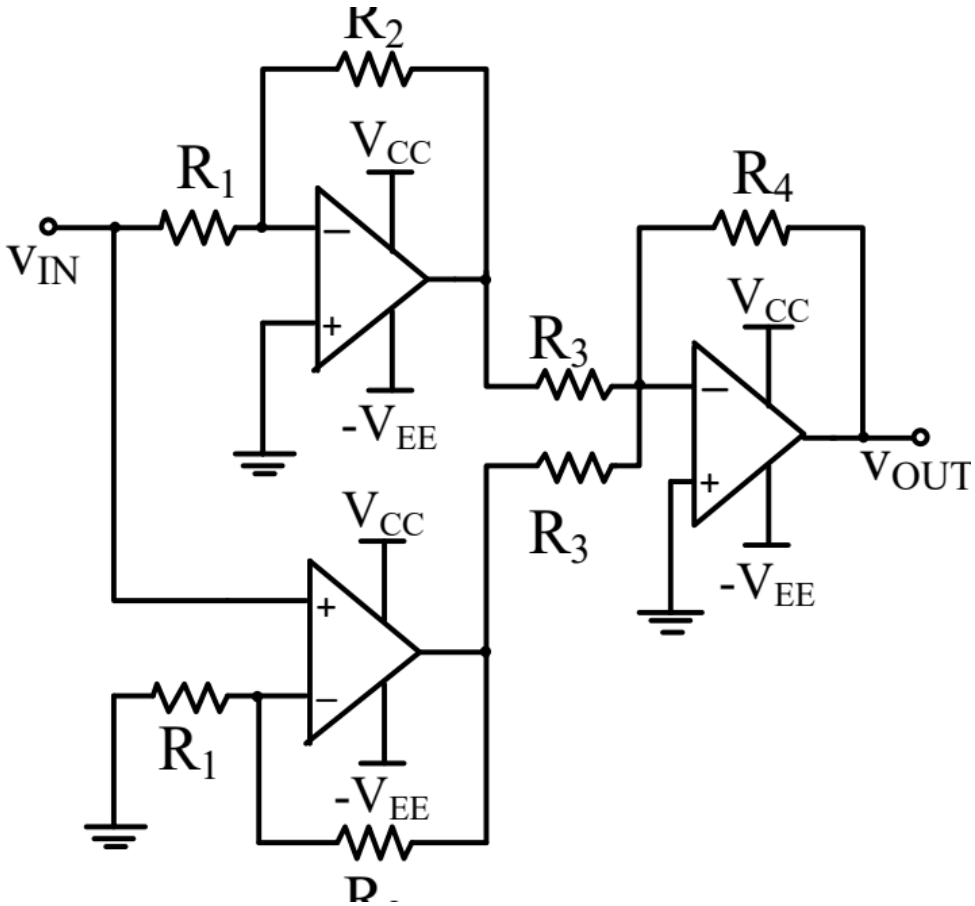
Answer: (A) [self-solved, no official key available]

Solution. Step 1. $\text{tr}(\mathbf{x}\mathbf{x}^T) = \sum_i x_i^2$.

Step 2. $= \mathbf{x}^T\mathbf{x} = l^2$.

GATE 2023 | Q.39 | 2 marks

The $\dfrac{V_{OUT}}{V_{IN}}$ of the circuit shown below is



(A) $-\dfrac{R_4}{R_3}$

(B) $\dfrac{R_4}{R_3}$

(C) $1 + \dfrac{R_4}{R_3}$

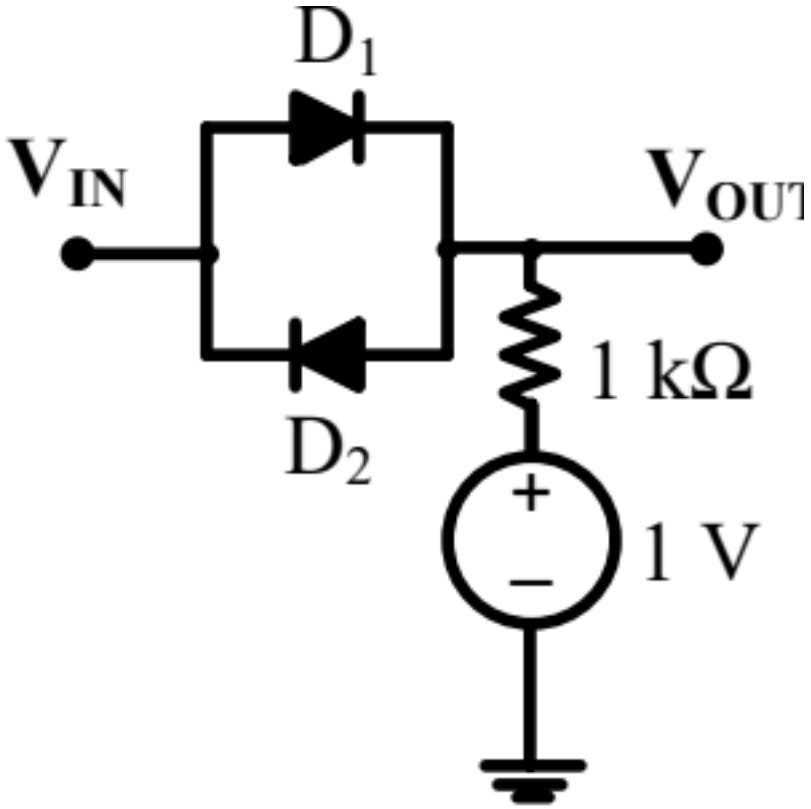
(D) $1 - \dfrac{R_4}{R_3}$

Answer: (B) [self-solved, no official key available]

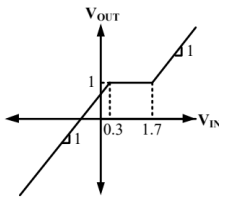
Solution. Step 1. The two input stages are matched buffers/amplifiers driven by the same V_{IN} in antiphase arrangement, so their gains contribute the same common factor.
Step 2. The final stage is a difference amplifier of gain R_4/R_3.
Step 3. $\dfrac{V_{OUT}}{V_{IN}} = \dfrac{R_4}{R_3}$.

GATE 2023 | Q.40 | 2 marks

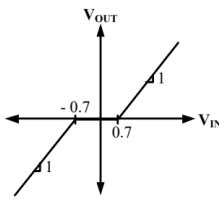
In the circuit shown below, D_1 and D_2 are silicon diodes with cut-in voltage of 0.7 V. V_{IN} and V_{OUT} are input and output voltages in volts. The transfer characteristic is



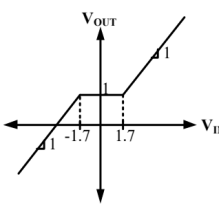
(A)



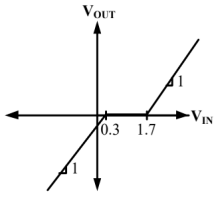
(B)



(C)



(D)



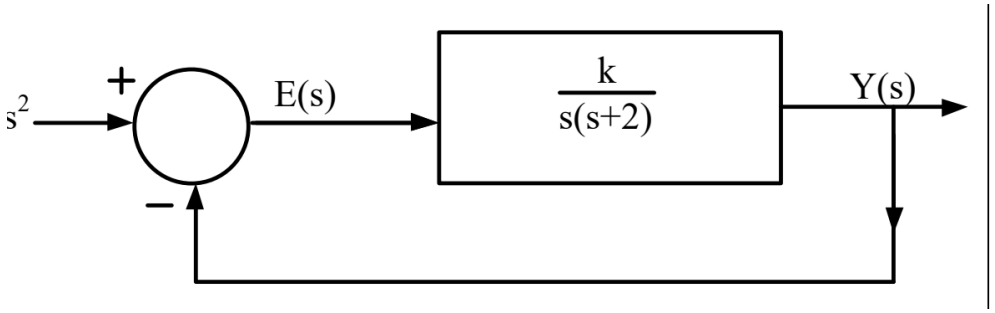
Answer: (A) [self-solved, no official key available]

Solution. Step 1. The antiparallel diodes conduct only if $|V_{IN} - V_{OUT}| > 0.7$ V; otherwise V_{OUT} is pulled to the 1 V source through the 1 kΩ resistor.
Step 2. For $0.3\,\text{V} < V_{IN} < 1.7$ V neither diode conducts: $V_{OUT} = 1$ V (flat).

Step 3. Outside this range the diodes conduct and $V_{OUT} = V_{IN} \mp 0.7$ (slope 1). This is plot (A).

GATE 2023 | Q.41 | 2 marks

A closed loop system is shown in the figure where $k > 0$ and $\alpha > 0$. The steady state error due to a ramp input ($R(s) = \alpha/s^2$) is given by



(A) $\frac{2\alpha}{k}$

(B) $\frac{\alpha}{k}$

(C) $\frac{\alpha}{2k}$

(D) $\frac{\alpha}{4k}$

Answer: (A) [self-solved, no official key available]

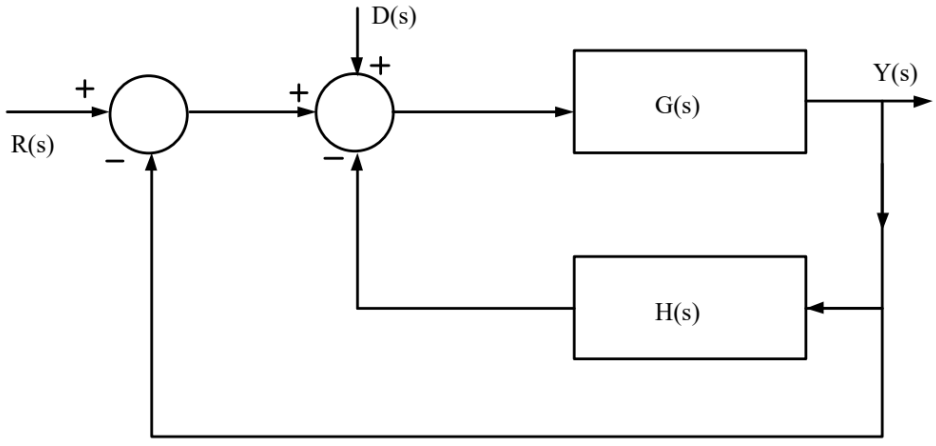
Solution. Step 1. The open loop transfer function is $G(s) = \frac{k}{s(s+2)}$ (type 1).

Step 2. $K_v = \lim_{s\to 0} sG(s) = \frac{k}{2}$.

Step 3. $e_{ss} = \frac{\alpha}{K_v} = \frac{2\alpha}{k}$.

GATE 2023 | Q.42 | 2 marks

In the following block diagram, $R(s)$ and $D(s)$ are two inputs. The output $Y(s)$ is expressed as $Y(s) = G_1(s)R(s) + G_2(s)D(s)$. $G_1(s)$ and $G_2(s)$ are given by



(A) $G_1 = \frac{G}{1+G+GH}$ and $G_2 = \frac{G}{1+G+GH}$

(B) $G_1 = \frac{G}{1+G+H}$ and $G_2 = \frac{G}{1+G+H}$

(C) $G_1 = \dfrac{G}{1+G+H}$ and $G_2 = \dfrac{G}{1+G+GH}$

(D) $G_1 = \dfrac{G}{1+G+GH}$ and $G_2 = \dfrac{G}{1+G+H}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Write the loop equation of the diagram and solve for Y in terms of R and D separately (superposition).

Step 2. Both transfer functions share the characteristic denominator $1+G+GH$, giving $G_1 = G_2 = \dfrac{G}{1+G+GH}$.

GATE 2023 | Q.43 | 2 marks

The state equation of a second order system is $\dot{\mathbf{x}}(t) = A\mathbf{x}(t)$, $\mathbf{x}(0)$ is the initial condition. Suppose λ_1 and λ_2 are two distinct eigenvalues of A and $\mathbf{v}_1$ and $\mathbf{v}_2$ are the corresponding eigenvectors. For constants α_1 and α_2, the solution, $\mathbf{x}(t)$, of the state equation is

(A) $\sum_{i=1}^{2} \alpha_i e^{\lambda_i t}\mathbf{v}_i$

(B) $\sum_{i=1}^{2} \alpha_i e^{2\lambda_i t}\mathbf{v}_i$

(C) $\sum_{i=1}^{2} \alpha_i e^{3\lambda_i t}\mathbf{v}_i$

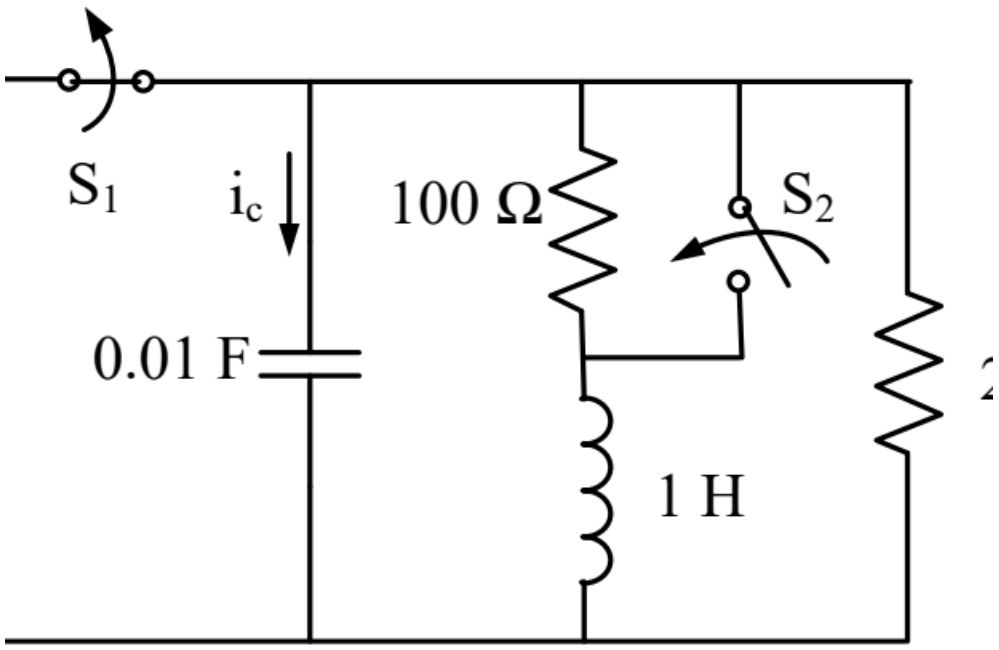
(D) $\sum_{i=1}^{2} \alpha_i e^{4\lambda_i t}\mathbf{v}_i$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Each eigenvector gives a mode: if $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ then $\mathbf{x} = e^{\lambda_i t}\mathbf{v}_i$ satisfies $\dot{\mathbf{x}} = A\mathbf{x}$.
Step 2. By linearity the general solution is $\sum \alpha_i e^{\lambda_i t}\mathbf{v}_i$.

GATE 2023 | Q.44 | 2 marks

The switch S_1 was closed and S_2 was open for a long time. At $t = 0$, switch S_1 is opened and S_2 is closed, simultaneously. The value of $i_c(0^+)$, in amperes, is



(A) 1

(B) -1

(C) 0.2
(D) 0.8

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Before switching the source current 1 A flows through S_1 and the circuit has reached dc steady state.
Step 2. At $t = 0^+$ the capacitor voltage and inductor current are unchanged. With S_1 open, the 1 A that was flowing is carried by the inductor and the capacitor must supply it through the new loop.
Step 3. Hence $i_c(0^+) = -1$ A.

GATE 2023 | Q.45 | 2 marks

Let a frequency modulated (FM) signal $x(t) = A\cos\left(\omega_c t + k_f \int_{-\infty}^{t} m(\lambda)d\lambda\right)$, where $m(t)$ is a message signal of bandwidth W. It is passed through a non-linear system with output $y(t) = 2x(t) + 5(x(t))^2$. Let B_T denote the FM bandwidth. The minimum value of ω_c required to recover $x(t)$ from $y(t)$ is

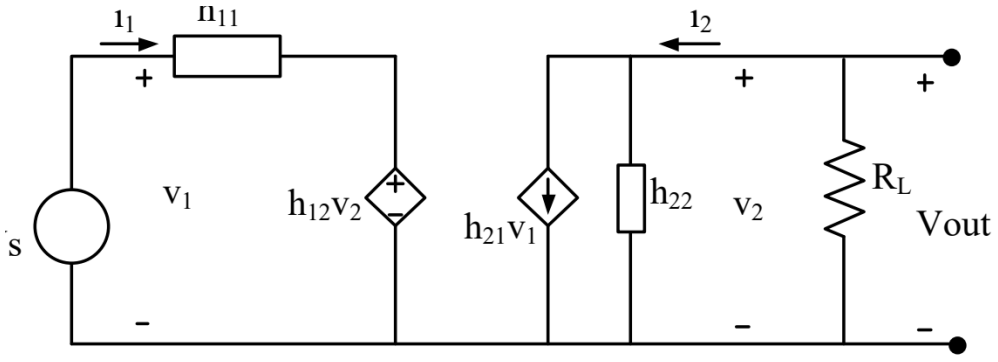
(A) $B_T + W$
(B) $\frac{3}{2}B_T$
(C) $2B_T + W$
(D) $\frac{5}{2}B_T$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. x^2 has a baseband component of width B_T (around 0) and a component around $2\omega_c$.
Step 2. The wanted term $2x(t)$ occupies $\omega_c \pm B_T/2$. To avoid overlap with the baseband term: $\omega_c - \frac{B_T}{2} \geq B_T$.
Step 3. $\omega_{c,min} = \frac{3}{2}B_T$.

GATE 2023 | Q.46 | 2 marks

The h-parameters of a two port network are shown below. The condition for the maximum small signal voltage gain $\frac{V_{out}}{V_s}$ is



(A) $h_{11} = 0,\ h_{12} = 0,\ h_{21}$ = very high and $h_{22} = 0$
(B) h_{11} = very high, $h_{12} = 0,\ h_{21}$ = very high and $h_{22} = 0$
(C) $h_{11} = 0,\ h_{12}$ = very high, h_{21} = very high and $h_{22} = 0$
(D) $h_{11} = 0,\ h_{12} = 0,\ h_{21}$ = very high and h_{22} = very high

Answer: (A) [self-solved, no official key available]

Solution. Step 1. With $h_{12} = 0$ there is no feedback; with $h_{11} = 0$ the whole source voltage appears at the input ($v_1 = V_s$).

Step 2. The output current $h_{21}i_1$ with $h_{22} = 0$ flows entirely through R_L. Hence $A_v = -\dfrac{h_{21}R_L}{h_{11}}$ is maximised for $h_{11} \to 0$, h_{21} very high.

GATE 2023 | Q.47 | 2 marks

Consider a discrete-time periodic signal with period $N = 5$. Let the discrete-time Fourier series (DTFS) representation be $x[n] = \sum_{k=0}^{4} a_k e^{jk\frac{2\pi n}{5}}$, where $a_0 = 1$, $a_1 = 3j$, $a_2 = 2j$, $a_3 = -2j$ and $a_4 = -3j$. The value of the sum $\sum_{n=0}^{4} x[n] \sin\dfrac{4\pi n}{5}$ is

(A) -10
(B) 10
(C) -2
(D) 2

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $\sin\dfrac{4\pi n}{5} = \dfrac{e^{j2\frac{2\pi n}{5}} - e^{-j2\frac{2\pi n}{5}}}{2j}$.
Step 2. $\sum_n x[n]e^{-jk\frac{2\pi n}{5}} = 5a_k$, so $\sum x[n]e^{j2\frac{2\pi n}{5}} = 5a_{-2} = 5a_3 = -10j$ and $\sum x[n]e^{-j2\frac{2\pi n}{5}} = 5a_2 = 10j$.
Step 3. Sum $= \dfrac{-10j - 10j}{2j} = -10$.

GATE 2023 | Q.48 | 2 marks

Let an input $x[n]$ having discrete time Fourier transform $X(e^{j\Omega}) = 1 - e^{-j\Omega} + 2e^{-3j\Omega}$ be passed through an LTI system. The frequency response of the LTI system is $H(e^{j\Omega}) = 1 - \dfrac{1}{2}e^{-j2\Omega}$. The output $y[n]$ of the system is

(A) $\delta[n] + \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$
(B) $\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$
(C) $\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$
(D) $\delta[n] + \delta[n-1] + \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. $Y = XH = \left(1 - e^{-j\Omega} + 2e^{-3j\Omega}\right)\left(1 - \frac{1}{2}e^{-2j\Omega}\right)$.
Step 2. Expanding: $1 - e^{-j\Omega} - \frac{1}{2}e^{-2j\Omega} + \frac{1}{2}e^{-3j\Omega} + 2e^{-3j\Omega} - e^{-5j\Omega}$.
Step 3. $y[n] = \delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$.

GATE 2023 | Q.49 | 2 marks

Let $x(t) = 10\cos(10.5Wt)$ be passed through an LTI system having impulse response $h(t) = \pi\left(\dfrac{\sin Wt}{\pi t}\right)^2 \cos 10Wt$. The output of the system is

(A) $\left(\dfrac{15W}{4}\right)\cos(10.5Wt)$
(B) $\left(\dfrac{15W}{2}\right)\cos(10.5Wt)$
(C) $\left(\dfrac{15W}{8}\right)\cos(10.5Wt)$
(D) $(15W)\cos(10.5Wt)$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $\dfrac{\sin Wt}{\pi t} \leftrightarrow$ rect of height 1 on $|\omega| < W$; its square transforms to a triangle: $\pi\left(\dfrac{\sin Wt}{\pi t}\right)^2 \leftrightarrow$

$W\left(1-\dfrac{|\omega|}{2W}\right)$ for $|\omega|<2W$.

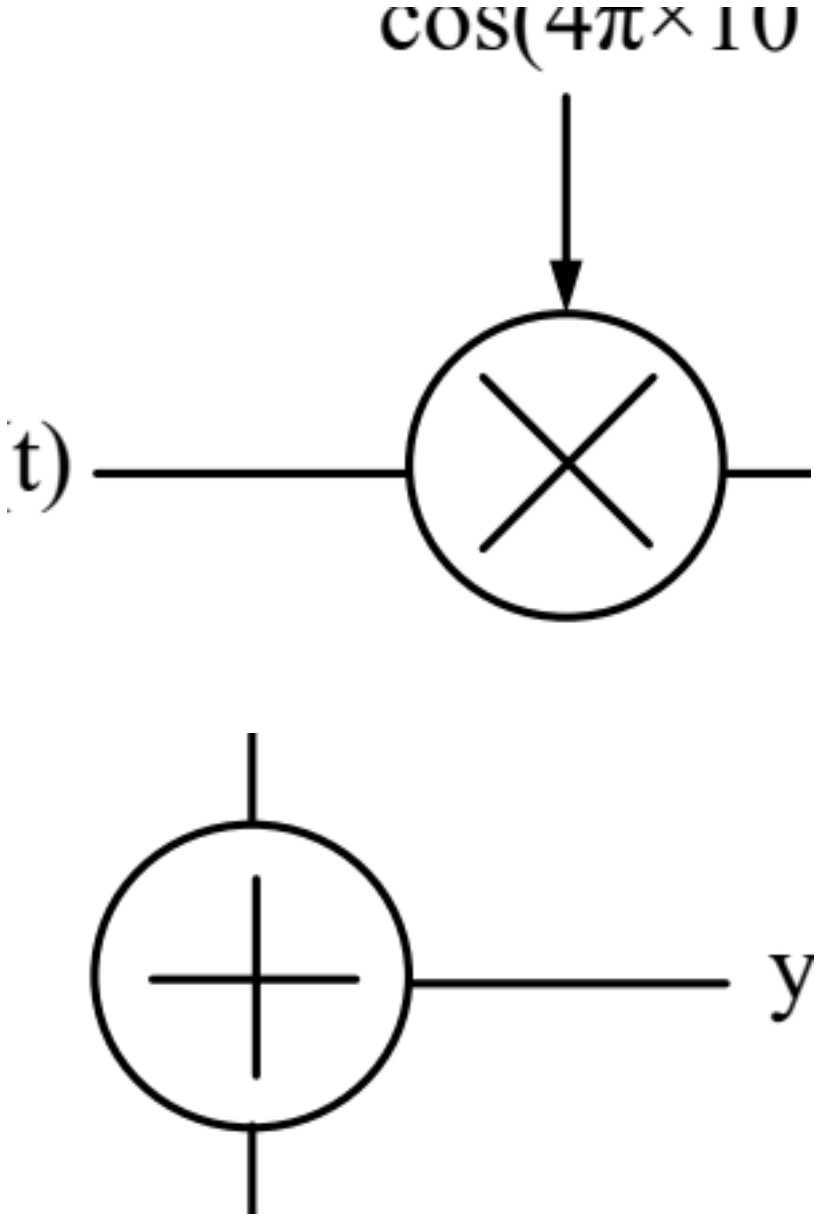
Step 2. Multiplying by $\cos 10Wt$ shifts it to $\pm 10W$ with factor $\frac{1}{2}$: $H(\omega)=\frac{1}{2}T(\omega\mp 10W)$.

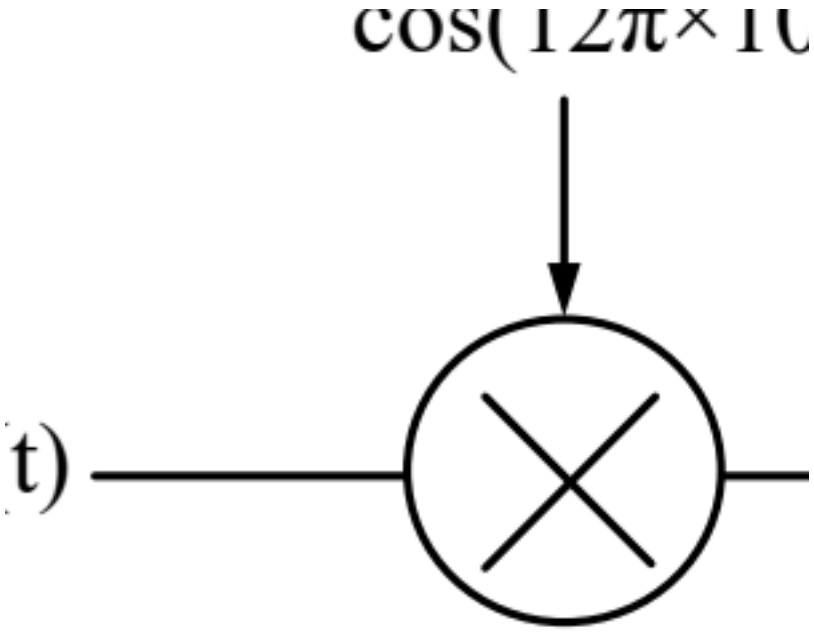
Step 3. At $\omega=10.5W$: $|H|=\frac{1}{2}W\left(1-\frac{0.5W}{2W}\right)=\frac{1}{2}\cdot 0.75W=\frac{3W}{8}$.

Step 4. Output amplitude $=10\times\dfrac{3W}{8}=\dfrac{15W}{4}$.

GATE 2023 | Q.50 | 2 marks

Let $x_1(t)$ and $x_2(t)$ be two band-limited signals having bandwidth $B=4\pi\times 10^3$ rad/s each. In the figure below, the Nyquist sampling frequency, in rad/s, required to sample $y(t)$, is





(A) $20\pi \times 10^3$
(B) $40\pi \times 10^3$
(C) $8\pi \times 10^3$
(D) $32\pi \times 10^3$

Answer: (D) [self-solved, no official key available]

Solution. Step 1. x_1 is multiplied by $\cos(4\pi \times 10^3 t)$: spectrum up to $4\pi \times 10^3 + 4\pi \times 10^3 = 8\pi \times 10^3$.
Step 2. x_2 is multiplied by $\cos(12\pi \times 10^3 t)$: spectrum up to $12\pi \times 10^3 + 4\pi \times 10^3 = 16\pi \times 10^3$ rad/s.
Step 3. The highest frequency in $y(t)$ is $16\pi \times 10^3$ rad/s, so the Nyquist rate is $2 \times 16\pi \times 10^3 = 32\pi \times 10^3$ rad/s.

GATE 2023 | Q.51 | 2 marks

The S-parameters of a two port network is given as $[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$ with reference to Z_0. Two lossless transmission line sections of electrical lengths $\theta_1 = \beta l_1$ and $\theta_2 = \beta l_2$ are added to the input and output ports for measurement purposes, respectively. The S-parameters $[S']$ of the resultant two port network is

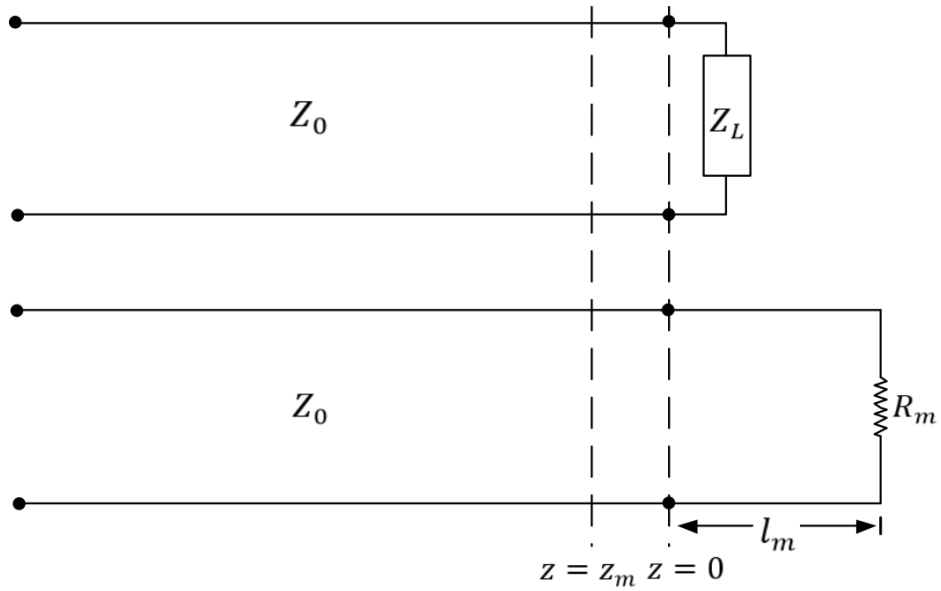
(A) $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$
(B) $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
(C) $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
(D) $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. A reflection measured through a line of electrical length θ travels the line twice: $S'_{11} = S_{11}e^{-j2\theta_1}$ and $S'_{22} = S_{22}e^{-j2\theta_2}$.
Step 2. A transmitted wave crosses both lines once: $S'_{12} = S_{12}e^{-j(\theta_1+\theta_2)}$, $S'_{21} = S_{21}e^{-j(\theta_1+\theta_2)}$.

GATE 2023 | Q.52 | 2 marks

The standing wave ratio on a 50 Ω lossless transmission line terminated in an unknown load impedance is found to be 2.0. The distance between successive voltage minima is 30 cm and the first minimum is located at 10 cm from the load. Z_L can be replaced by an equivalent length l_m and terminating resistance R_m of the same line. The value of R_m and l_m, respectively, are



(A) $R_m = 100\ \Omega,\ l_m = 20$ cm
(B) $R_m = 25\ \Omega,\ l_m = 20$ cm
(C) $R_m = 100\ \Omega,\ l_m = 5$ cm
(D) $R_m = 25\ \Omega,\ l_m = 5$ cm

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Minima are $\lambda/2$ apart: $\lambda/2 = 30$ cm, $\lambda = 60$ cm.

Step 2. At a voltage minimum the impedance is purely resistive: $R_m = \dfrac{Z_0}{\text{SWR}} = \dfrac{50}{2} = 25\ \Omega$.

Step 3. Replace the load by a line of length l_m terminated in R_m: the first minimum from the load is at $30 - l_m = 10$ cm (since minima repeat every 30 cm), so $l_m = 20$ cm.

GATE 2023 | Q.53 | 2 marks

The electric field of a plane electromagnetic wave is $\mathbf{E} = \mathbf{a}_x C_{1x}\cos(\omega t - \beta z) + \mathbf{a}_y C_{1y}\cos(\omega t - \beta z + \theta)$ V/m. Which of the following combination(s) will give rise to a left handed elliptically polarized (LHEP) wave?

(A) $C_{1x} = 1,\ C_{1y} = 1,\ \theta = \pi/4$
(B) $C_{1x} = 2,\ C_{1y} = 1,\ \theta = \pi/2$
(C) $C_{1x} = 1,\ C_{1y} = 2,\ \theta = 3\pi/2$
(D) $C_{1x} = 2,\ C_{1y} = 1,\ \theta = 3\pi/4$

Answer: (A), (B), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. For a wave travelling in $+z$, E_y leading E_x in phase ($0 < \theta < \pi$) gives left-handed polarisation; $\pi < \theta < 2\pi$ gives right-handed.

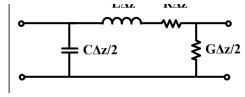
Step 2. The wave is elliptical unless $C_{1x} = C_{1y}$ and $\theta = \pm\pi/2$.

Step 3. (A) $\theta = \pi/4$: LH elliptical. (B) $\theta = \pi/2$, unequal amplitudes: LH elliptical. (C) $\theta = 3\pi/2$: right-handed. (D) $\theta = 3\pi/4$: LH elliptical.

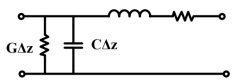
GATE 2023 | Q.54 | 2 marks

The following circuit(s) representing a lumped element equivalent of an infinitesimal section of a transmission line is/are

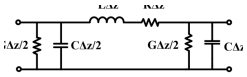
(A)



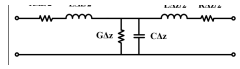
(B)



(C)



(D)

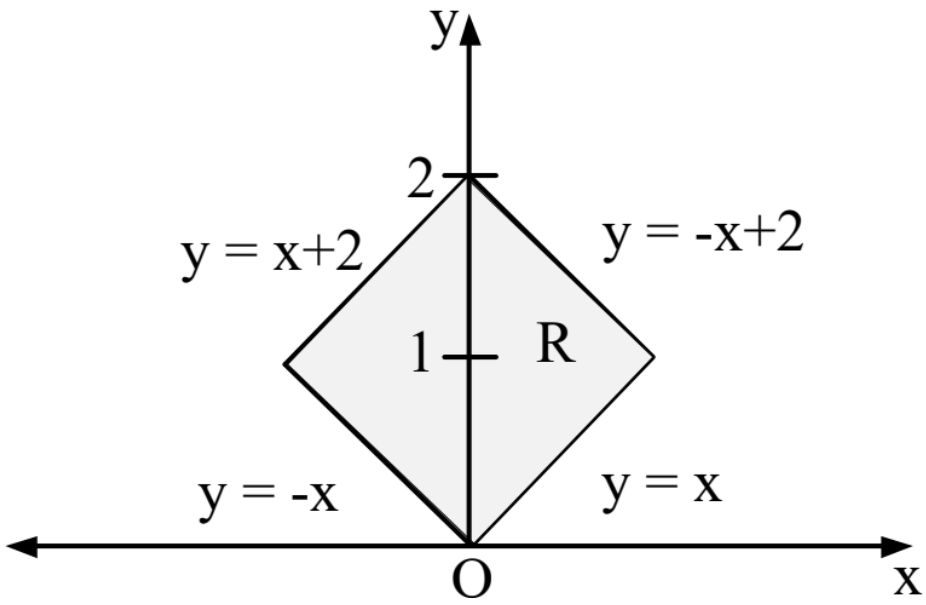


Answer: (A), (B), (C), (D) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. A transmission-line section of length Δz is modelled by series $R\Delta z$, $L\Delta z$ and shunt $G\Delta z$, $C\Delta z$.
Step 2. The shunt elements may be placed at either end or split half and half (T and π forms); as $\Delta z \to 0$ all arrangements reduce to the same telegrapher's equations, so all four circuits are valid.

GATE 2023 | Q.55 | 2 marks

The value of the integral $\iint_R xy\,dx\,dy$ over the region R, given in the figure, is _____ (rounded off to the nearest integer).



Answer: 0 [self-solved, no official key available]

Solution. Step 1. The region (a square with vertices $(0,0),(1,1),(0,2),(-1,1)$) is symmetric about the line $x=0$.
Step 2. The integrand xy is odd in x, so the contributions from $x>0$ and $x<0$ cancel: the integral is 0.

GATE 2023 | Q.56 | 2 marks

In an extrinsic semiconductor, the hole concentration is given to be $1.5n_i$ where n_i is the intrinsic carrier concentration of 1×10^{10} cm^{-3}. The ratio of electron to hole mobility for equal hole and electron drift current is given as _____ (rounded off to two decimal places).

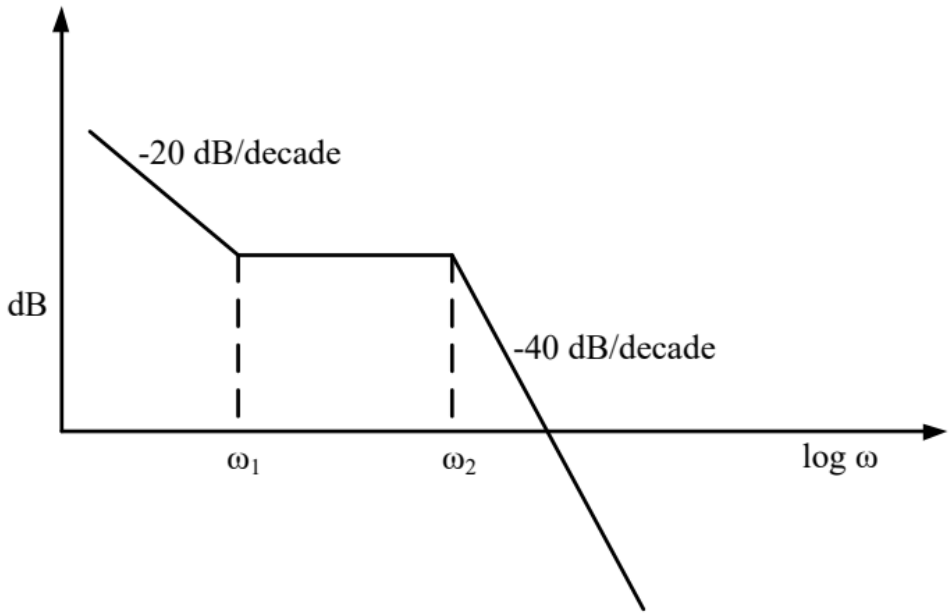
Answer: 2.25 [self-solved, no official key available]

Solution. Step 1. $n = \dfrac{n_i^2}{p} = \dfrac{n_i}{1.5}$.

Step 2. Equal drift currents: $qn\mu_n E = qp\mu_p E \Rightarrow \dfrac{\mu_n}{\mu_p} = \dfrac{p}{n} = \dfrac{1.5n_i}{n_i/1.5} = 2.25$.

GATE 2023 | Q.57 | 2 marks

The asymptotic magnitude Bode plot of a minimum phase system is shown in the figure. The transfer function of the system is $G(s) = \dfrac{k(s+z)^a}{s^b(s+p)^c}$, where k, z, p, a, b and c are positive constants. The value of $(a+b+c)$ is _____ (rounded off to the nearest integer).

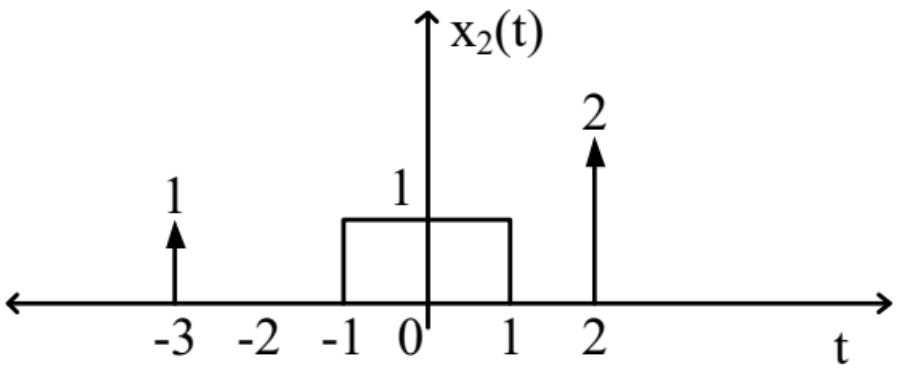


Answer: 4 [self-solved, no official key available]

Solution. Step 1. Initial slope -20 dB/decade $\Rightarrow$ a single pole at the origin: $b = 1$.
Step 2. At ω_1 the slope returns to 0: a zero, $+20$ dB/dec, so $a = 1$.
Step 3. At ω_2 the slope becomes -40 dB/dec, i.e. change of $-40 \Rightarrow$ two poles: $c = 2$.
Step 4. $a + b + c = 1 + 1 + 2 = 4$.

GATE 2023 | Q.58 | 2 marks

Let $x_1(t) = u(t+1.5) - u(t-1.5)$ and $x_2(t)$ is shown in the figure below. For $y(t) = x_1(t) * x_2(t)$, the $\displaystyle\int_{-\infty}^{\infty} y(t)\,dt$ is _____ (rounded off to the nearest integer).



Answer: 12 [self-solved, no official key available]

Solution. Step 1. For convolution, $\int y\,dt = \left(\int x_1\,dt\right)\left(\int x_2\,dt\right)$.
Step 2. $\int x_1\,dt = 3$ (a pulse of height 1 and width 3).

Step 3. $\int x_2\,dt = 1 + 1 + 2 = 4$ (from the impulses and the pulse in the figure), so the integral is $3 \times 4 = 12$.

GATE 2023 | Q.59 | 2 marks

Let $X(t)$ be a white Gaussian noise with power spectral density $\frac{1}{2}$ W/Hz. If $X(t)$ is input to an LTI system with impulse response $e^{-t}u(t)$, the average power of the system output is _____ W (rounded off to two decimal places).

Answer: 0.25 [self-solved, no official key available]

Solution. Step 1. Output power $= \frac{N_0}{2}\int_{-\infty}^{\infty}|h(t)|^2 dt$ with $\frac{N_0}{2} = \frac{1}{2}$.

Step 2. $\int_0^{\infty} e^{-2t}dt = \frac{1}{2}$, so $P = \frac{1}{2}\cdot\frac{1}{2} = 0.25$ W.

GATE 2023 | Q.60 | 2 marks

A transparent dielectric coating is applied to glass ($\varepsilon_r = 4$, $\mu_r = 1$) to eliminate the reflection of red light ($\lambda_0 = 0.75$ μm). The minimum thickness of the dielectric coating, in μm, that can be used is _____ (rounded off to two decimal places).

Answer: 0.13 [self-solved, no official key available]

Solution. Step 1. For an anti-reflection coating the coating index is $n_c = \sqrt{n_{air}n_{glass}} = \sqrt{1\cdot 2} = \sqrt{2}$ and its thickness is a quarter wavelength in the coating.

Step 2. $d = \frac{\lambda_0}{4n_c} = \frac{0.75}{4 \times 1.414} = 0.13$ μm.

GATE 2023 | Q.61 | 2 marks

In a semiconductor device, the Fermi-energy level is 0.35 eV above the valence band energy. The effective density of states in the valence band at $T = 300$ K is 1×10^{19} cm^{-3}. The thermal equilibrium hole concentration in silicon at 400 K is _____ $\times10^{13}$ cm^{-3} (rounded off to two decimal places). Given kT at 300 K is 0.026 eV.

Answer: 63.49 [self-solved, no official key available]

Solution. Step 1. $p = N_V\, e^{-(E_F - E_V)/kT}$ with $N_V \propto T^{3/2}$.

Step 2. $N_V(400\text{ K}) = 10^{19}\left(\frac{400}{300}\right)^{3/2} = 1.54 \times 10^{19}$ cm^{-3} and $kT = 0.026 \times \frac{400}{300} = 0.0347$ eV.

Step 3. $p = 1.54 \times 10^{19}\, e^{-0.35/0.0347} = 6.35 \times 10^{14} = 63.49 \times 10^{13}$ cm^{-3}.

GATE 2023 | Q.62 | 2 marks

A sample and hold circuit is implemented using a resistive switch and a capacitor with a time constant of 1 μs. The time for the sampling switch to stay closed to charge a capacitor adequately to a full scale voltage of 1 V with 12-bit accuracy is _____ μs (rounded off to two decimal places).

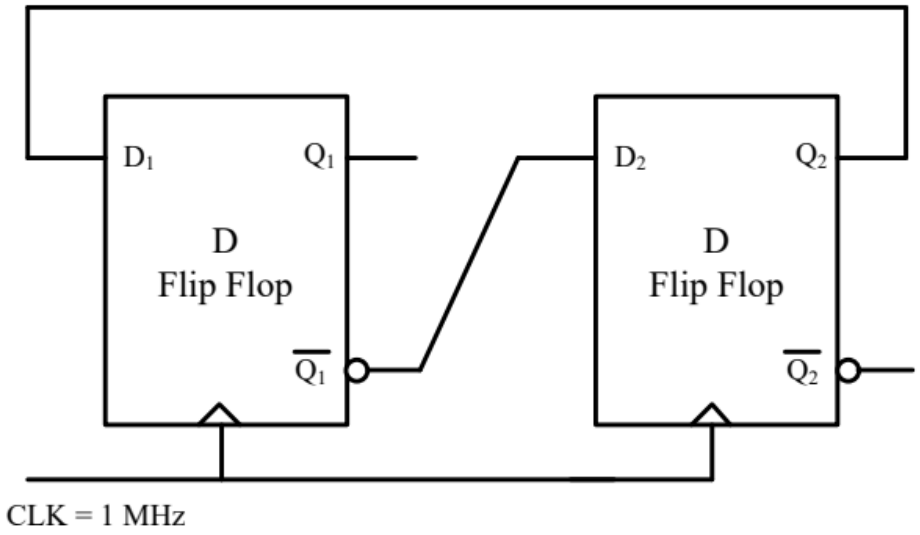
Answer: 8.32 [self-solved, no official key available]

Solution. Step 1. The residual error after time t is $e^{-t/\tau}$ (fraction of full scale). For 12-bit accuracy it must be $\leq 2^{-12}$.

Step 2. $t = \tau \ln(2^{12}) = 12\ln 2 \times 1$ μs $= 8.32$ μs.

GATE 2023 | Q.63 | 2 marks

In a given sequential circuit, initial states are $Q_1 = 1$ and $Q_2 = 0$. For a clock frequency of 1 MHz, the frequency of signal Q_2 in kHz, is _____ (rounded off to the nearest integer).

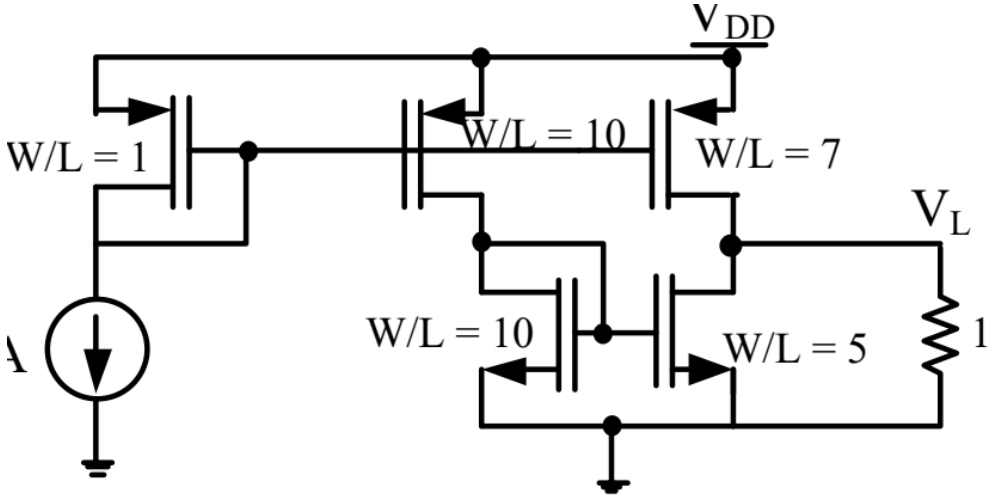


Answer: 250 [self-solved, no official key available]

Solution. Step 1. The two D flip-flops form a twisted-ring (Johnson) counter: $D_1 = \overline{Q_2}$, $D_2 = Q_1$.
Step 2. The sequence (Q_1Q_2): $10 \to 11 \to 01 \to 00 \to 10$: period of 4 clocks.
Step 3. $f_{Q_2} = \dfrac{1 \text{ MHz}}{4} = 250$ kHz.

GATE 2023 | Q.64 | 2 marks

In the circuit below, the voltage V_L is _____ V (rounded off to two decimal places).



Answer: 2.00 [self-solved, no official key available]

Solution. Step 1. The 1 mA reference flows in the diode-connected device of $W/L = 1$, which is mirrored to the devices of $W/L = 7$ (top) and $W/L = 10$ (lower-left).
Step 2. The $W/L = 7$ device sources 7 mA into the load node. The 10 mA branch drives a lower diode-connected device of $W/L = 10$, mirrored to the $W/L = 5$ device that sinks 5 mA from the node.
Step 3. Net current into the 1 kΩ resistor $= 7 - 5 = 2$ mA, so $V_L = 2.00$ V.

GATE 2023 | Q.65 | 2 marks

The frequency of occurrence of 8 symbols (a–h) is shown in the table below. A symbol is chosen and it is determined by asking a series of "yes/no" questions which are assumed to be truthfully answered. The average number of questions when asked in the most efficient sequence, to determine the chosen symbol, is _____ (rounded off to two decimal places).

Symbols	a	b	c	d	e	f	g	h
Frequency of occurrence	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{128}$

Answer: 1.98 [self-solved, no official key available]

Solution. Step 1. The most efficient question sequence corresponds to a Huffman (optimal prefix) code; the average number of questions equals the average code length.
Step 2. The probabilities are dyadic $(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots, \frac{1}{128}, \frac{1}{128})$, so the average length equals the entropy: $H = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{16}(4) + \frac{1}{32}(5) + \frac{1}{64}(6) + 2 \cdot \frac{1}{128}(7) = 1.984$.
Step 3. Answer ≈ 1.98.